Best Practices for Stored Procedures

Presented by Kevin Kline
SQL Server MVP since 2004

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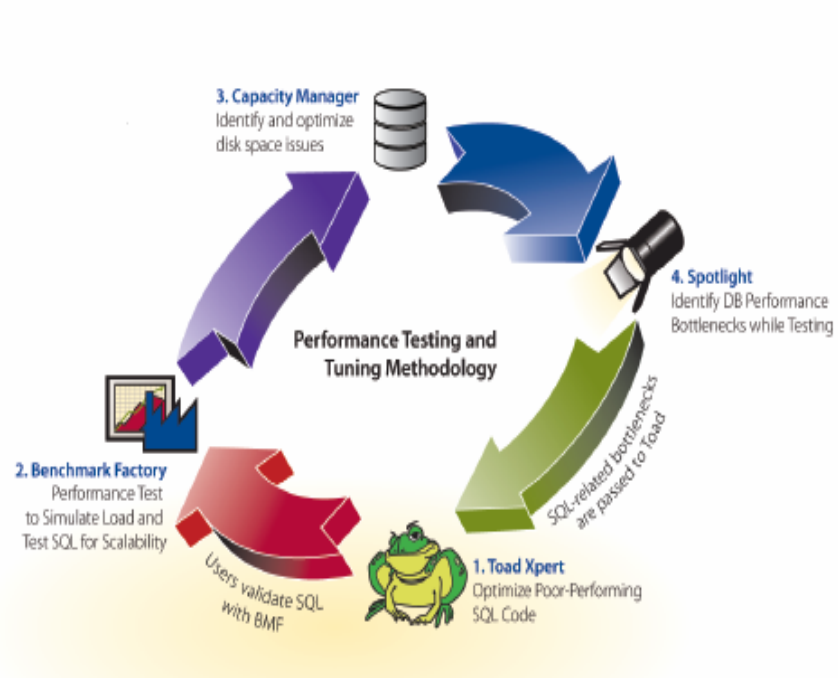
Agenda

- ➔ **Speaker Bio**
- **About Quest Software**
- **Accolades & Awards**
- **Stored Procedure Best Practices**
- **Call to Action**
- **Q & A**



Kevin Kline

- *Technical Strategy Manager for SQL Server Solutions,* Quest Software Inc.
- Microsoft SQL Server MVP
- President of the International SQL Server User's Group – PASS
- SQL Server expert and author of the O'Reilly titles "SQL in a Nutshell" and "Transact-SQL Programming"

Interactive Audience Poll

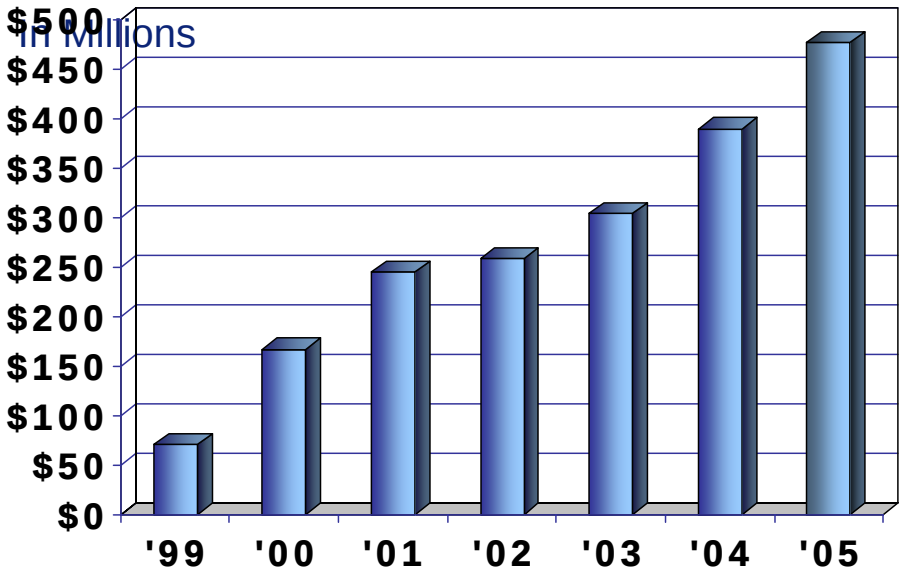
- What is your primary job duty?
  - DBA, Dev, BI, Architect, IT Pro, Mgr, Other?
  - Helps me tailor the content
- When will you use SQL Server 2005?
  - Already using it, 1-3 months, 4-6 months, More than 6 months but less than one year, Later
- How much SQL Server 2005 are you using?
  - No production systems on SQL2005, 1 or 2 production systems on SQL2005, as much as 50% on SQL2005, 100%!
- When will you use 64-bit platforms?
  - Already using it, 1-3 months, 4-6 months , More than 6 months but less than one year, Later

Speaker Bio – Kevin Kline

- Started in IT in 1986. BS in MIS in 1989 from University of Alabama.
- Microsoft SQL Server MVP since 2004
- Author of 7 database books
  - 1 on Oracle, 2 on SQL Server, 2 on SQL, 1 on DB design, 1 on DB benchmarking
- President of PASS (www.sqlpass.org)
  - Conference is next Sept in Denver, CO
  - Over 130 sessions on SQL Server, BI & Dev
- Blogs for SQLMag.com and SQLBlog.com
- Writes monthly columns for Database Trends & Applications and SQL Server Magazine
- Worked for NASA, US Army, and Deloitte & Touche before Quest Software.

About Quest Software

- Profitable & Growing
  - Over a half-billion in ’06 revenues
  - Continued growth in ’01 through ‘06 against sector trends
- Founded in 1987
- Public in 1999; Nasdaq: QSFT
- Customers: 18,000 Worldwide
  - Including 75 percent of the Fortune 500
- over 2,750 employees
  - Over 950 in R&D
  - Nearly a dozen R&D centers around the globe
- Headquarters: Aliso Viejo, CA



Accolades & Awards

- Best of Tech Ed award 2006 with Spotlight on SQL Server
- SQL Server Magazine, Platinum reader's choice award
- SQL Server Magazine Readers Choice Awards, winner in 11 categories, 2005
- No. 1 vendor in Distributed Data Management Facilities, IDC, 2005
- Microsoft ISV Partner of the Year, 2004
- Microsoft TechEd Best of Show Winner, 2004
- Microsoft TechEd Europe Best of Show Winner, 2005
- No. 1 vendor in Application Management Software, Gartner Dataquest, 2005
- Jolt Productivity Award Winner
- Network Computing Editor's Choice Winner
- No. 8 in the "Who's Who in Enterprise Software," Investor's Business Daily

• Toad™ for SQL Server Xpert
• Knowledge Xpert
• Toad™ for SQL Server Xpert
• Litespeed for SQL Server Developer Edition
• SQL Turbo
• Capacity Manager
• Toad™ for SQL Server Xpert
• Benchmark Factory
• Data Factory
• Quest Comparison Suite
• Quest Comparison Suite / Toad for SQL Server (DBA Module)
• LiteSpeed™ for SQL Server
• Toad™ for SQL Server
• Toad™ Data Modeler
DELIVER
Plan
Develop
Optimize
Validate
Deploy
Solutions for SQL Server
Administer
Inform
Resolve
Diagnose
Discover
MANAGE
• Toad™ for SQL Server (DBA Module)
• LiteSpeed™ for SQL Server
• Foglight® Cartridge for SQL Server
• Spotlight® on SQL Server Enterprise
• Capacity Manager
• Benchmark Factory®
• SQL Watch
• Toad™ for SQL Server
• SQL Watch
• Quest Comparison Suite
• Capacity Manager
• Toad™ Data Modeler
• Capacity Manager
• Toad™ for SQL Server Xpert
• Spotlight® on SQL Server Enterprise
• Database Analysis (Quest Central®)
• Spotlight® on SQL Server Enterprise
• Performance Analysis (Quest Central®)
• Coefficient

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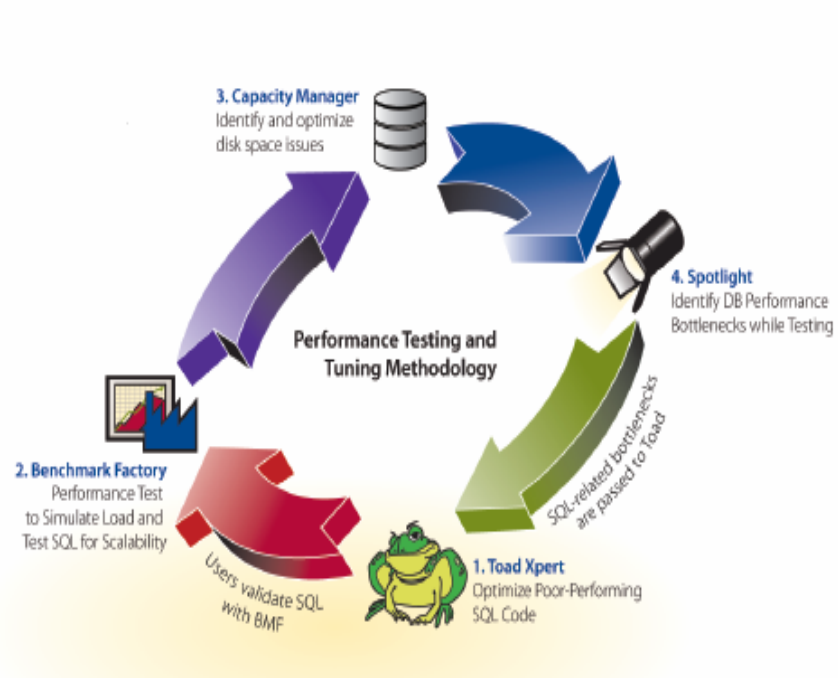


Naming Procs

- Do NOT use sp_xxx as a naming convention.
- Causes additional searches and added I/O.
  - SQL Server will scan the procedure cache for Master, no matter what database the procedure was executed from
  - SQL Server will then acquire an exclusive COMPILE lock to perform a second search in the local database
  - If a user stored procedure has same name as an sp_xxx stored procedure in MASTER, then the user procedure will NEVER be used.

Calling Procs

- Use stored procedure calls rather than embedded SQL
- EXEC versus SP_EXECUTESQL
  - same behavior with regard to batches, the scope of names, and database context
  - EXEC compiles entire SQL at one time
  - SP_EXECUTE compiles and executes as an execution plan separate from the execution plan of the batch that called **sp_executesql** itself.
  - SP_EXECUTESQL executes a Transact-SQL statement or batch that can be reused many times, or that has been built dynamically, using a single execution plan.
  - Often better than EXEC.

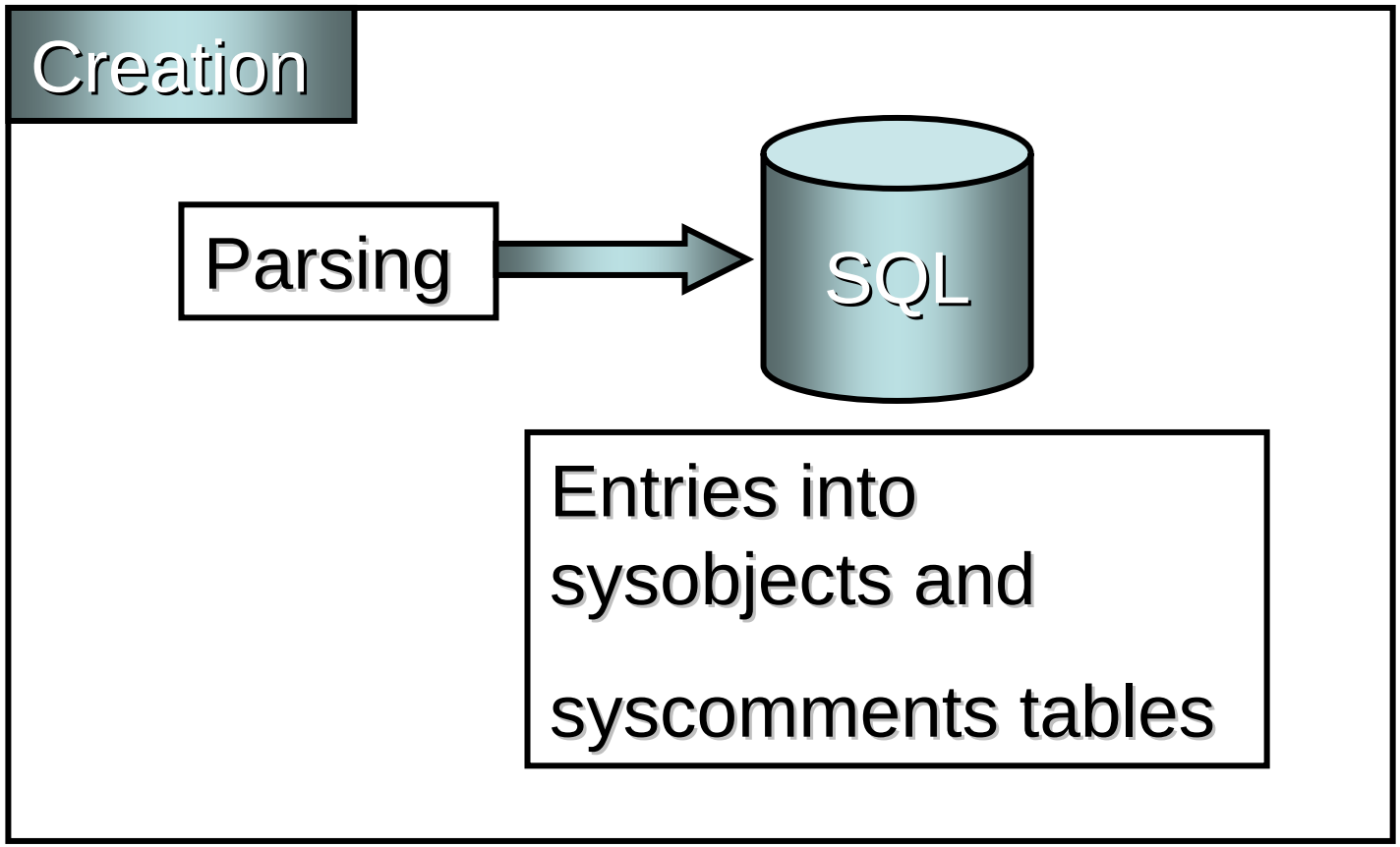
Common Problems

- Un-necessary network communication
- Execution Plans
  - Not Enough Sharing
  - Too Much Sharing
  - Inadequate plan
- Recompilation
- Transactions
- Row Theory when Set Theory is better
- Other Delays

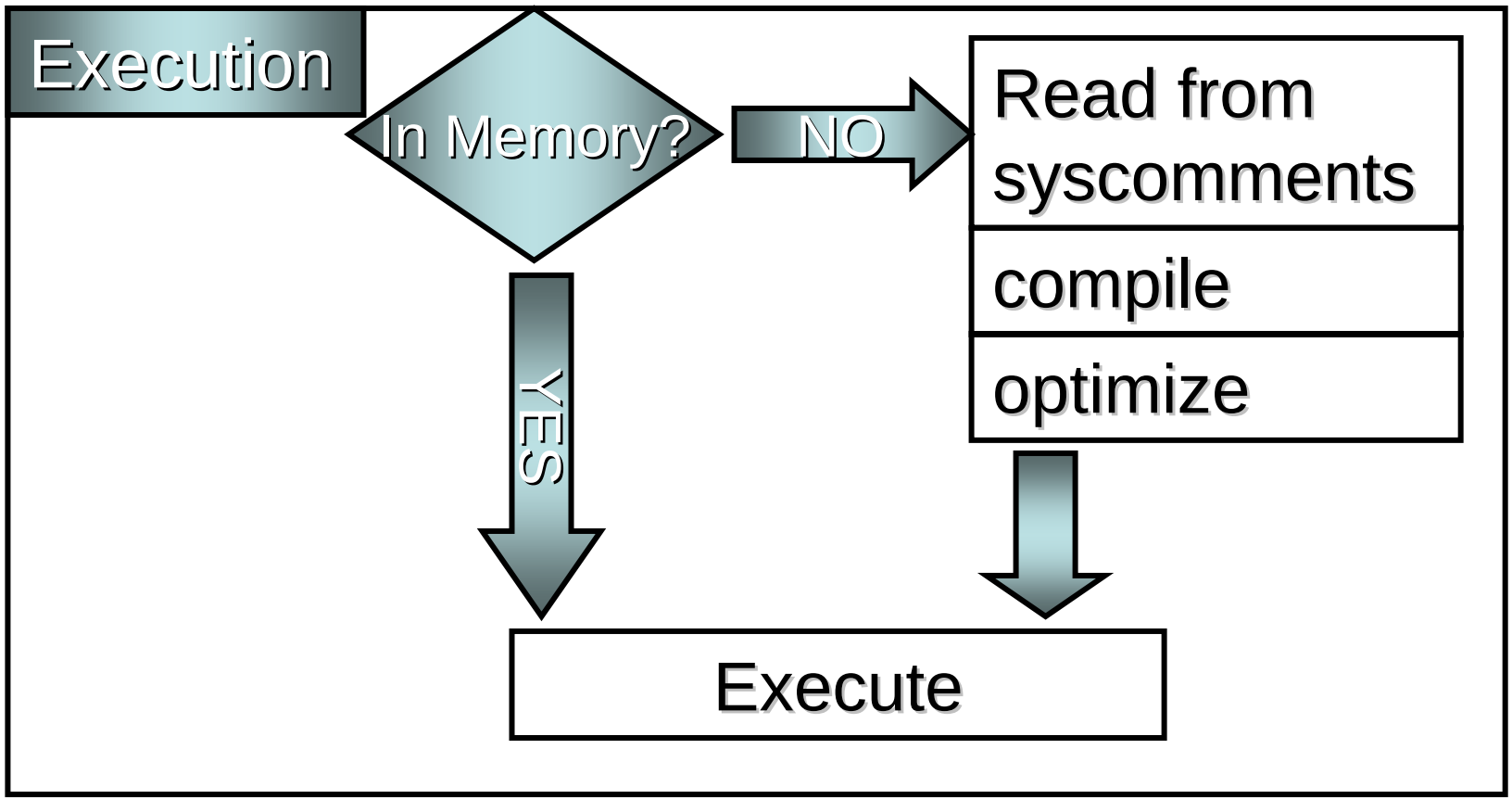
Set Nocount On

- No done_in_proc messages
  - <10 rows affected>
  - Extra network trip
- Server setting via Trace 3640
  - -T3640
  - sp_configure ‘user options’, 512

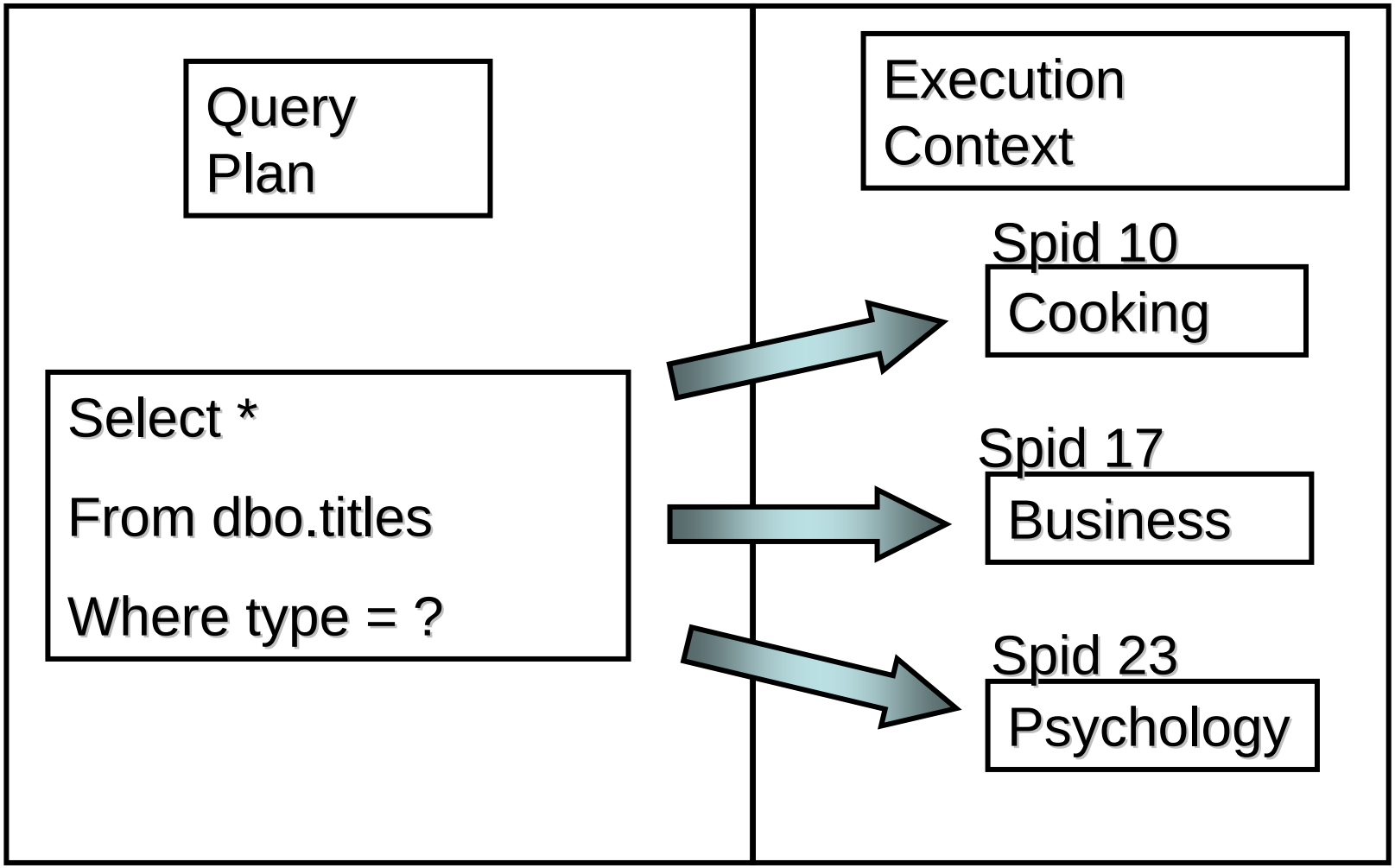
Create Stored Procedure



Exec Stored Procedure



SP Plan Sharing



Not Enough Plan Sharing

- Set Options
- Language used by the login
- Dateformat for the connection

	ON	OFF	Setable By:	
Arithabort	▽		S,D,A	
Concat_null_yields_null	▽		S,D,A	
Quoted_Identifier**	▽		S,D,A	
Ansi_nulls**	▽▲		S,D,A	
Ansi_Padding	▽		S,D,A	
Ansi_Warnings	▽▲		S,D,A	
Numeric_roundabort		▽	S,D,A	
Forceplan			SET only	
Ansi_Null_Dflt_On			S,D,A	
Ansi_Null_Dflt_Off			S,D,A	

▽ = required for index views or computed columns

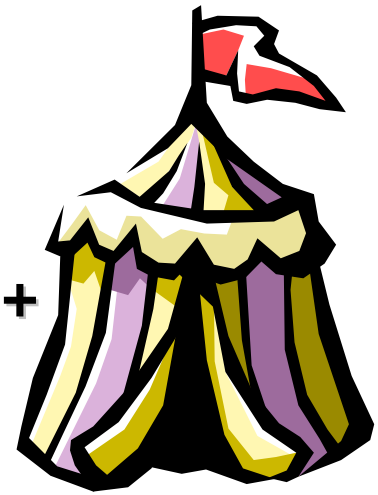
▲ = required for distributed queries

S, D, A = sp_configure, sp_dboption, alter database

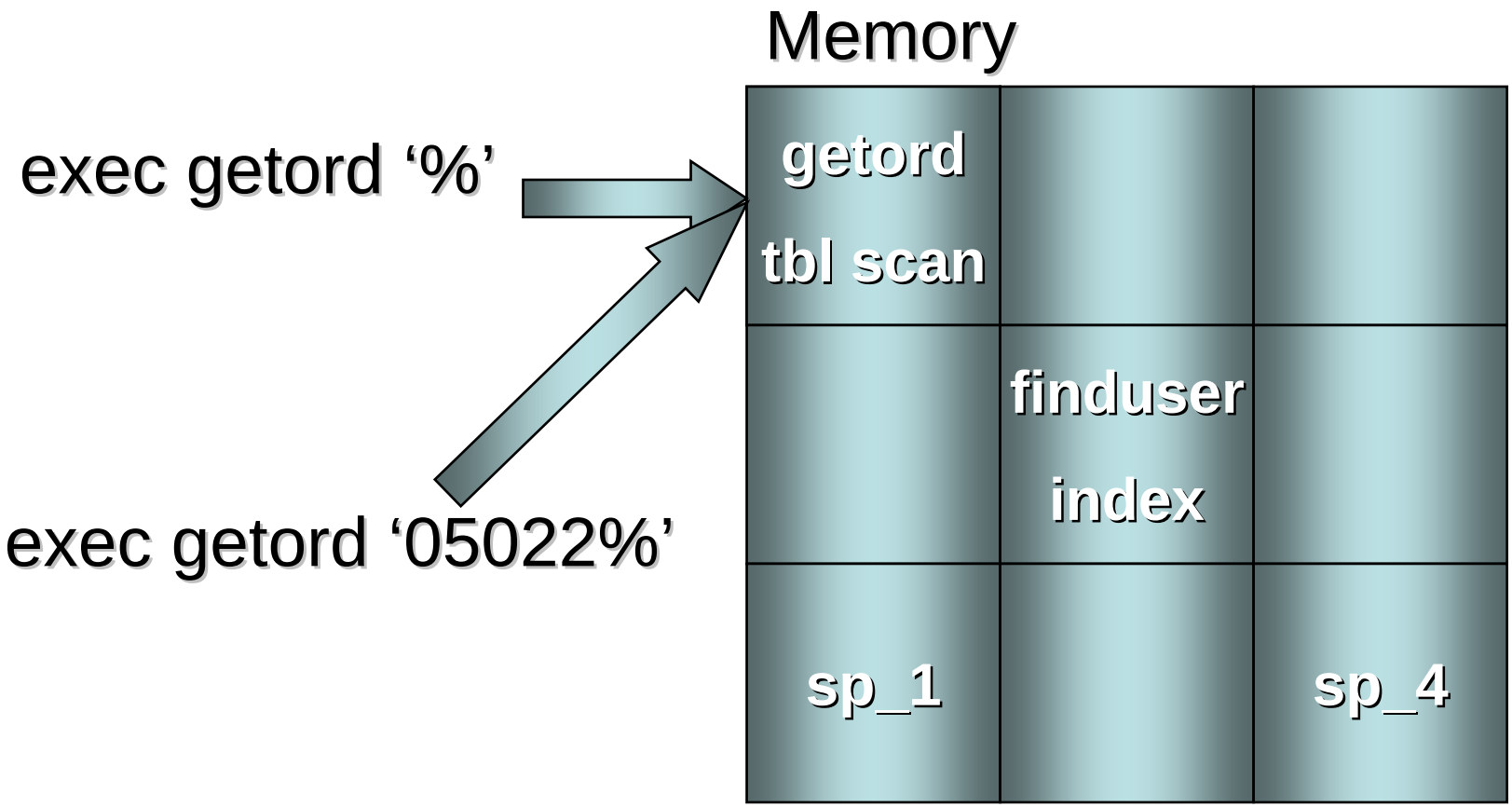
Option Hierarchy

- Set Statement in code
- OLEDB/ODBC Connection string
- ODBC
  - Control Panel
  - SQLConfigDatasource
- OLEDB/ODBC auto set of 7
  - DB-Library/Embedded SQL for C-NOT
- Database Level Settings (alter database)
- Server Wide Settings (sp_configure)

+ **SIS**

+ 

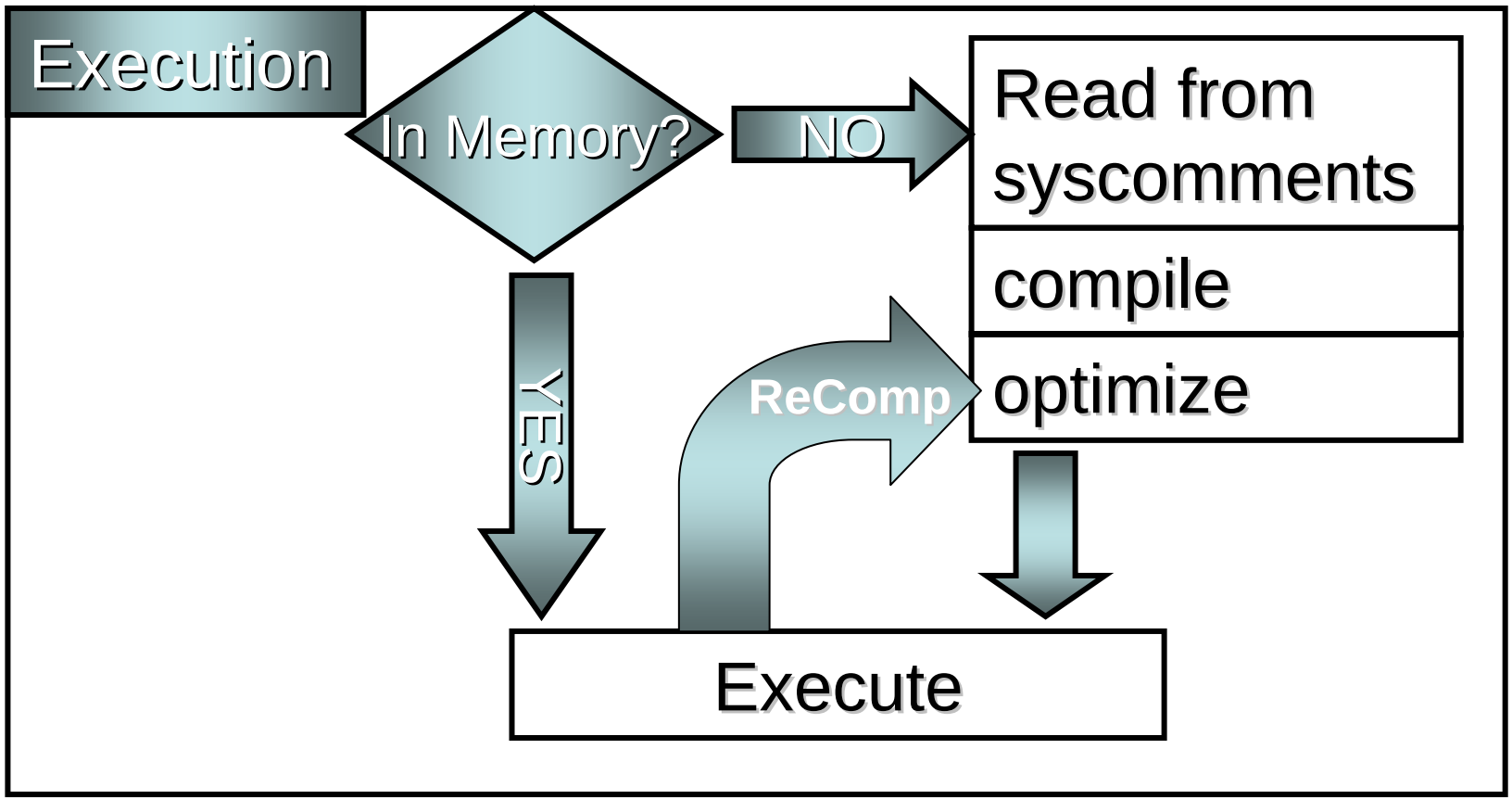
Too Much Plan Sharing



Monitoring Plan Caching

- DBCC FreeProcCache
- DBCC FlushProcInDB(<dbid>)
- DBCC DropCleanBuffers
- Actual Query Plan
- SQL Profiler
- Syscacheobjects

Too Many Recompiles



Proc Recompiles

- Because we request it
- Previous plan aged out of memory
- Interleaved DDL and DML
- Schema changes to objects in Proc
- New index statistics
- Cursor on temp table (SQL 7.0)
- Sp_configure

Requested Recompile

- Create proc … with recompile as
- Exec myproc … with recompile
- sp_recompile titles

Proc Plan Aging

Memory

getord 16 12		
	finduser 7 4	

Interleaved DDL and DML

```
create proc testDDLDML as
    create table testdml    (DDL)
    insert into testdml     (DML – RECOMPILE)
    alter table testdml     (DDL)
    insert into testdml     (DML – RECOMPILE)
    drop table testdml
```

Schema Changes to Objects

- Sysobjects.schema_ver
  - Column additions, deletions
  - Data type changes
  - Constraint additions, deletions
  - Rule/Default bindings
- Query plan index dropped
- Related concept: always declare columns
  - Avoid using SELECT * FROM foo
  - Especially avoid INSERT without declared columns

New Index Statistics

- Auto_update statistics
- Auto_create statistics
- Update statistics

Inner proc Recompiles

- When it uses the outer procs temp table
  - First time (if called multi times)

Using Wildcards in WHERE clauses with Params

- Found in constructs like:

```
SELECT *
FROM foo
WHERE emp_ID = '%'
```

- Optimizer guesses percentage of rows returned

Operator	%
=	10
>	30
<	30
Between	10

Transactions

- Avoid nested transactions. They aren't truly nested:
  - COMMIT only saves data from the outermost transaction
  - ROLLBACK nukes *ALL* transactions, both innermost and outermost
- Orphaned Transactions
  - Errors don't usually abort a transaction except for deadlocks
  - Returning from a procedure or batch does NOT commit a transaction. Once a transaction is started, it remains open until:
    - The transaction is committed
    - The transaction is rolled back
    - The connection ends
- Use @@TRANCOUNT or sys.dm_tran_active_transactions to look for orphaned transactions when entering a new routine
- Keep transactions as short as possible!
- Keep transactions explicit!
- Remember lock escalation!

Row Theory when Set Theory is Better

- C#, VB, VB.NET, etc are procedural languages where you tell the program what to do
- SQL is a declarative language where you tell it what you want
- SQL becomes inefficient when you tell it what to do using CURSORs or LOOPs
- Learn to think in *sets*!
  - One pass through a table or tables
  - Let the computer process joins and filters

Best Practices

- Owner qualify all names (2-part)
- Standardize user options, language
- Minimize use of tempdb
  - Use table variable or smart structures instead
- Help the optimizer cache plans
- Avoid using wildcards with parameters in WHERE clause
- Don’t interleave DDL and DML
- TEST,TEST, TEST

dbo.sptest

select * from test

(Kev) Exec sptest

dbo.test

dbo stuff

Kev.test

Kev stuff

Resources

- SQL Server Books Online
  - “Execution Plan Caching and Reuse”
- “Analyzing Optimal Compiled Plan Caching” - Sajal Dam
- Knowledge Base
  - Q243588, “INF: Troubleshooting Performance of Ad Hoc Queries”
  - Q243586, “INF: Troubleshooting Stored Procedure Recompilation”
  - Q263889, “INF: SQL Blocking Due to [COMPILE] locks”
- Inside SQL Server 2000 – Kalen Delaney
- Guru’s Guide to … - Ken Henderson

(Don’t forget the drawing)

Call to Action – Next Steps

- Attend a live demo, especially ***Spotlight on SQL Server*** and ***Performance Analysis***: http://www.quest.com/landing/qc_demos.asp
- Download white papers: http://www.quest.com/whitepapers
- Get a trial versions: http://www.quest.com/solutions/download.asp
- Email us with your questions: sales@quest.com or call 949.754.8000

DRAWING FOR GOODIES!

Questions & Answers

Thank you for attending this session and the

More questions? Email me at kevin.kline@quest.com